

Modelling Bowling “Strategy” with Hidden Markov Models

The idea

The k-means analysis (`analysis.md`) found each bowler’s repertoire of **delivery types** (clusters), based purely on the physical properties of the ball (speed, length, line, swing, seam, angles). That analysis treats every delivery independently — it has no notion of time or sequence. This document uses **exactly the same clusters** (the over/round-the-wicket split, e.g. Morkel’s O0 “Short”, R2 “Out-swinger”, etc., with the same expert-given names) — so a cluster name means the same thing in both documents.

In reality, a bowler doesn’t pick each ball at random. Across an over or a spell they tend to settle into a **plan** — e.g. “attack the stumps with the new ball” or “bowl tight and build pressure” — and that plan shapes which delivery types and outcomes show up in a run of consecutive balls. We can’t observe the plan directly, but we can observe its consequences: which cluster each ball belongs to, and what happened (dot ball / runs / wicket).

This is exactly the setup a **Hidden Markov Model (HMM)** is designed for (background reading):

- A small number of **hidden states** represent the bowler’s underlying “strategy” at each point in time.
- A **transition matrix** describes how likely the bowler is to stay in the same strategy from ball to ball, or switch to another.
- An **emission distribution** describes what we expect to observe (cluster + outcome) given the current hidden strategy.

Given a sequence of observations, we can fit these three pieces and then run the **Viterbi algorithm** to decode the most likely hidden-strategy sequence underlying each bowler’s deliveries.

Bayesian vs. simple fitting

The BayesHMM article fits this model fully Bayesian (Stan, MCMC, priors on every parameter). That’s the rigorous way to do it, but it’s a lot of machinery for ~350-560 observations per bowler. In keeping with “nothing fancy”, this analysis fits the **same model structure** (categorical HMM: discrete states, discrete observations) using the standard **Baum-Welch / EM algorithm** (`hmmlearn`’s `CategoricalHMM`), which gives maximum-likelihood estimates of the same π (initial probabilities), A (transition matrix) and θ (emission probabilities). The interpretation — filtering, smoothing, Viterbi decoding — is identical; we’re just using point estimates instead of posterior distributions.

Data preparation

For each of the four bowlers, every delivery is reduced to a single observed **symbol**:

symbol = (ball cluster) × (outcome)

- **Ball cluster**: the per-bowler over/round-the-wicket-split cluster from `analysis.md` (4–6 clusters depending on bowler, e.g. O0/O1/.../R0/R1/...), capturing *what kind of ball it was*. The HMM’s “alphabet size” (`n_clusters` × 3 outcomes) is therefore the same total cluster count as in `analysis.md`: Morkel 6, Broad 5, Starc 6, Cummins 4.
- **Outcome**: one of three categories capturing *what happened*:
 - **Dot** — no runs conceded, no wicket
 - **Runs** — runs conceded off the bat, no wicket
 - **Wicket** — the bowler took the wicket

Deliveries are ordered **chronologically** (match start date □ innings □ over □ ball), so the HMM sees each bowler’s deliveries in the order they were actually bowled, across all 51 matches in the dataset. The 7 tracking-error deliveries (<75 mph) excluded from the clustering are excluded here too.

Strategies reset each match

A bowler doesn’t carry a “strategy” across a gap of weeks or months between Test matches — whatever plan he had going into the last ball of one match has no bearing on the first ball of the next. Originally this analysis

concatenated all of a bowler's deliveries into one long sequence and let the HMM model transitions across match boundaries too, which implicitly assumes the strategy *can* carry over from the last ball of match N to the first ball of match N+1.

To fix this, each bowler's deliveries are still placed in one long chronological array (348–385 symbols, alphabet size $n_clusters \times 3 = 12\text{--}18$), but `hmmlearn` is given a `lengths` array marking where each of the bowler's matches starts and ends. This means:

- The **transition matrix** A is only ever applied *within* a match — there's no learned transition from the last ball of one match to the first ball of the next.
- The **start probabilities** π are applied at the start of *every* match, not just once at the start of the whole sequence — so π now has a real interpretation: “what state does this bowler tend to begin a match in?”
- The **emission distributions** are still shared and fit across all matches (pooling data for statistical power), but decoding (Viterbi) is run match-by-match.

This is a standard way to handle multiple independent sequences with a single HMM, and it directly answers the question of whether a bowler's “starting strategy” is consistent match-to-match (persistent) or essentially reset/random.

Choosing the number of hidden states

We fit the HMM separately for each bowler with **2, 3, 4, and 5 hidden states** (15 random restarts each, keeping the best-likelihood fit), and compare them with the **Bayesian Information Criterion (BIC)**, which penalises extra parameters — useful here because more states quickly add a lot of emission parameters relative to the amount of data.

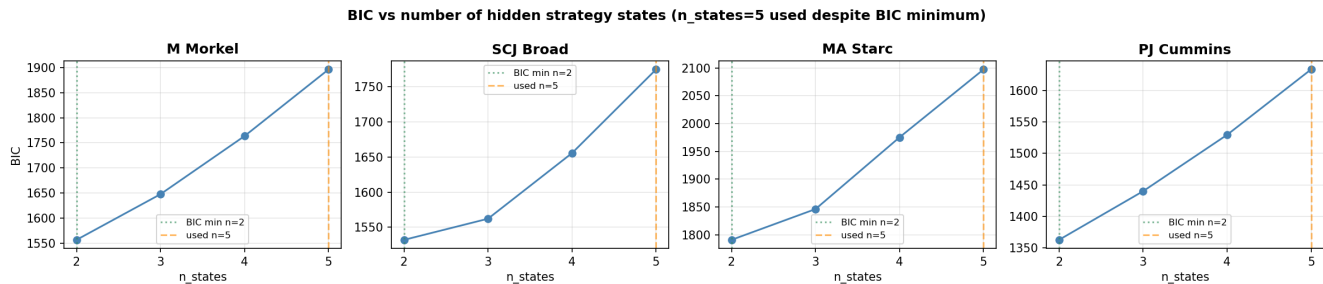


Figure 1: Choosing the number of hidden states

For **all four bowlers**, **BIC is minimised at 2 states** and rises steadily and substantially through 3, 4 and 5 (e.g. Morkel goes from 1556.8 at 2 states to 1896.2 at 5). Statistically, a two-state model is about as much structure as 350–560 balls per bowler can support — at 5 states, several states are decoded from only 10–20 balls, so they should be read as suggestive rather than well-estimated.

Despite that, the results below deliberately use 5 hidden states per bowler. The 2-state model (used in earlier drafts of this analysis) turned out to map almost exactly onto each bowler's **over/round-the-wicket split** — interesting, but it doesn't tell us much about *strategy* beyond “which side of the stumps”. The aim of pushing to 5 states is to see whether, underneath that dominant split, there's any finer-grained structure — e.g. different “modes” *within* a bowler's round-the-wicket spells, or a short-lived high-wicket “strike” state that a 2-state model would average away. As the BIC plot shows, this is a **deliberate trade of statistical parsimony for exploratory detail** — treat the small, rarely-occupied states (anything under ~10% occupancy) as a hint to investigate further with more data, not a confirmed finding.

States are relabelled (after fitting) in order of “aggression” (wicket rate $\times 10$ + runs rate) — State 0 is the most defensive, State 4 the most aggressive — purely so states can be compared consistently across the figures. This relabelling means State 0...State 4 here are not the “same” states as in the 2-state version of this analysis — the model has been refit from scratch with 5 components.

Results

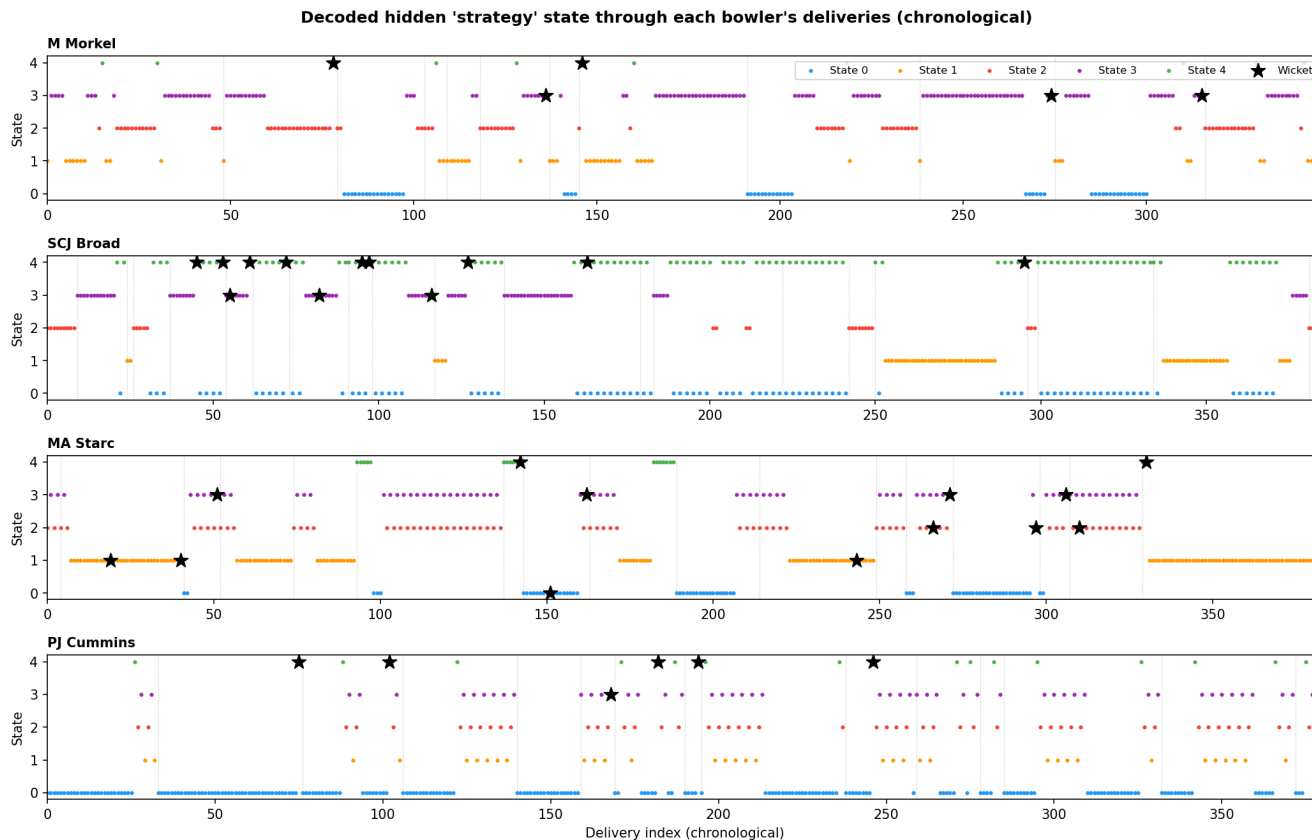


Figure 2: Decoded hidden state through time

The timeline plot shows the Viterbi-decoded state for every delivery in chronological order (★ = wicket); the thin grey dotted lines mark **match boundaries** — the HMM resets to its start distribution at each one, and no transition is modelled across them. The emission plot shows, for each state, the model's *fitted* probability of bowling each ball cluster (top row) and each outcome (bottom row). The histogram below it shows the same idea from the *decoded* side: for each bowler, it takes every delivery the Viterbi path actually assigned to State 0 / State 1, and plots what share of those deliveries were each cluster. Cluster codes (O0, O1, R0, ...) are the same ones used in `analysis.md` — see each bowler's table below for what they mean.

M Morkel

	State 0 (16.1%, ~11 balls)	State 1 (14.4%, ~3 balls)	State 2 (25.0%, ~7 balls)	State 3 (41.4%, ~6 balls)	State 4 (3.2%, ~1 ball)
Dominant clusters	R1 Full outside off (59%), R0 In-swinger (22%), R2 Out-swinger (13%)	R0 In-swinger (58%), R2 Out-swinger (38%)	O0 Short (27%), O1 Good length (73%)	R2 Out-swinger (48%), R1 Full outside off (22%), R0 In-swinger (19%), R3 Bouncer (10%)	O1 Good length (57%), O0 Short (21%), R2 Out-swinger (22%)
Outcome	Dot 95%, Runs 5%, Wicket 0%	Dot 93%, Runs 7%, Wicket 0%	Dot 79%, Runs 21%, Wicket 0%	Dot 60%, Runs 37%, Wicket 2%	Dot 22%, Runs 60%, Wicket 18%
Starts a match in this state	1 / 12	6 / 12	5 / 12	0 / 12	0 / 12

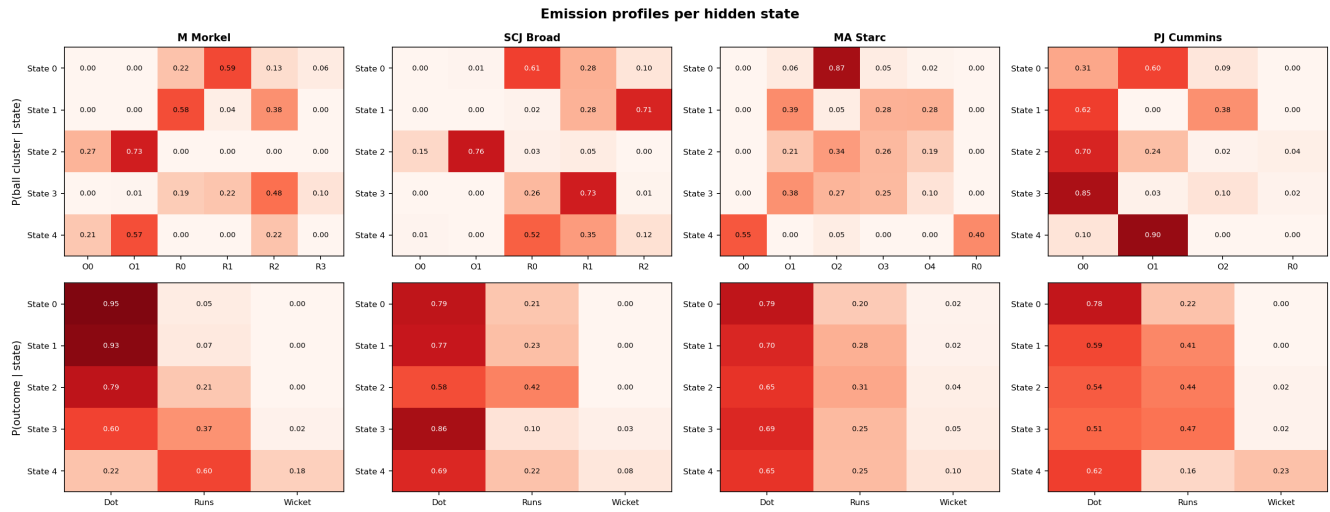


Figure 3: Emission profiles per hidden state

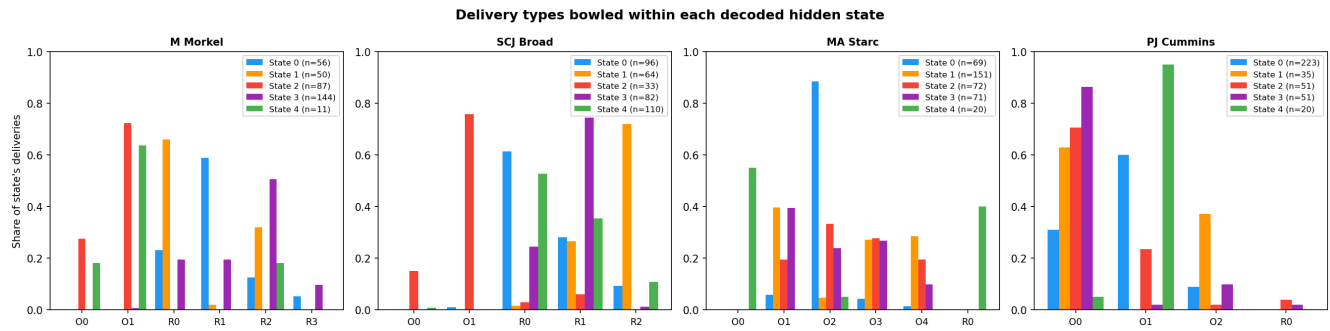


Figure 4: Delivery types bowled within each decoded hidden state

The over/round-the-wicket split is still the dominant axis — States 0, 1 and 3 are essentially 100% **round-the-wicket** clusters, State 2 is essentially 100% **over-the-wicket** (O0/O1) — but at 5 states the round-the-wicket repertoire now splits into **three sub-modes of increasing “leakiness”**: State 0 (full outside off-heavy, almost nothing scored, Dot 95%), State 1 (in-swinger/out-swinger mix, Dot 93%), and State 3 — by far the most-occupied state overall (41.4%) — where the same out-swinger/in-swinger mix concedes far more freely (Runs 37%) and produces his only “normal” wickets (2%). State 2 (over-the-wicket short/good length) sits in between (Dot 79%). Finally, **State 4 is a tiny (3.2% ≈ 11 balls) mixed Over/Round state with a striking Wicket rate of 18%** — the model’s transition matrix shows State 2 occasionally drops into it (13% of the time) before bouncing straight back to State 1. Given it represents only ~11 deliveries, this should be read as “something interesting may be happening here” rather than a confirmed pattern — but it’s exactly the kind of short-lived, high-impact state that a 2-state model would average into the background.

SCJ Broad

	State 0 (24.9%, ~1 ball)	State 1 (16.6%, ~14 balls)	State 2 (8.6%, ~9 balls)	State 3 (21.3%, ~8 balls)	State 4 (28.6%, ~1 ball)
Dominant clusters	R0 Back of length, nip-away (61%), R1 Full and straight (28%), R2 Good length, big swing away (10%)	R2 Good length, big swing away (71%), R1 Full and straight (28%)	O1 Good length, angle across (76%), O0 Short ball/bouncer (15%)	R1 Full and straight (73%), R0 Back of length, nip-away (26%)	R0 Back of length, nip-away (52%), R1 Full and straight (35%), R2 Good length, big swing away (12%)
Outcome	Dot 79%, Runs 21%, Wicket 0%	Dot 77%, Runs 23%, Wicket 0%	Dot 58%, Runs 42%, Wicket 0%	Dot 86%, Runs 10%, Wicket 3%	Dot 69%, Runs 22%, Wicket 8%
Starts a match in this state	0 / 21	2 / 21	5 / 21	5 / 21	9 / 21

Broad’s deliveries are overwhelmingly **round-the-wicket** (States 0, 1, 3, 4 are all 99–100% round-the-wicket clusters); only State 2 (8.6%) is a genuinely over-the-wicket excursion, and it’s also his leakiest non-wicket-taking state (Dot 58%, Runs 42%). Within his round-the-wicket bowling, the model now distinguishes a **persistent containment mode** (State 1, ~14-ball dwell, swing-away heavy, no wickets), a **tight full-and-straight mode** (State 3, Dot 86% — his most defensive state — but still carrying a 3% wicket rate), and a pair of **rapidly-alternating modes** (States 0 and 4, both ~1-ball dwell, transitioning into each other almost every ball — see the transition matrix: State 0 → State 4 always, State 4 → State 0 90% of the time). Of that oscillating pair, **State 4 carries by far his highest wicket rate (8%)** despite a very similar cluster mix to State 0 (which has 0% wickets) — both are dominated by the same back-of-length/full-and-straight/swing-away trio, but the model is picking up some other signal (perhaps sequencing or build-up) that distinguishes the two. As with Morkel’s State 4, this 8%-wicket state is built from a meaningful chunk of data (28.6% of deliveries) so is more trustworthy than a 1-ball curiosity, but the *mechanism* distinguishing it from its near-identical neighbour (State 0) isn’t obvious from the cluster mix alone.

MA Starc

	State 0 (18.0%, ~11 balls)	State 1 (39.4%, near-absorbing)	State 2 (18.8%, ~1 ball)	State 3 (18.5%, ~1 ball)	State 4 (5.2%, ~9 balls)
Dominant clusters	O2 Length, in-swinger (87%)	O1 Length, seam in (39%), O3 Full in-swinger (28%), O4 Bouncer (28%)	O2 Length, in-swinger (34%), O3 Full in-swinger (26%), O1 Length, seam in (21%), O4 Bouncer (19%)	O1 Length, seam in (38%), O2 Length, in-swinger (27%), O3 Full in-swinger (25%), O4 Bouncer (10%)	O0 Length, out-swinger (55%), R0 Round-the-wicket length (40%)
Outcome	Dot 79%, Runs 20%, Wicket 2%	Dot 70%, Runs 28%, Wicket 2%	Dot 65%, Runs 31%, Wicket 4%	Dot 69%, Runs 25%, Wicket 5%	Dot 65%, Runs 25%, Wicket 10%
Starts a match in this state	6 / 17	0 / 17	7 / 17	1 / 17	3 / 17

Starc bowls almost entirely **over the wicket** — only State 4 (5.2%) mixes in any round-the-wicket deliveries (40%), so the over/round split barely registers for him at all. State 0 reprises the **tight, repetitive in-swinger-length opening** (O2, 87%, Dot 79%) seen at 2 states. State 1 is again **near-absorbing** (dwell ~142 balls — once Starc settles into this broad seam-in/full-in-swinger/bouncer mix, he stays there for the rest of the spell) and is by far his most-bowled state (39.4%). The new structure at 5 states is States 2 and 3: a pair of **ball-to-ball alternating states** (State 2 $\square$ State 3 92% of the time, State 3 $\square$ State 2 always) that split what was previously lumped into “the broad mix” into two slightly different blends — both noticeably more wicket-prone (4–5%) than States 0/1 (2%). **State 4 — the same out-swinger/round-the-wicket-length mix flagged in the 2-state version — again stands out with a 10% wicket rate**, the highest of any state for Starc. *Caveat:* as before, `analysis.md` notes O0 (Length, out-swinger) contains a handful of deliveries with an anomalous (physically implausible) positive Drop Angle reading — a Hawk-Eye tracking artefact, not a genuine delivery type — and these sit inside State 4.

PJ Cummins

	State 0 (58.7%, ~10 balls)	State 1 (9.2%, ~1 ball)	State 2 (13.4%, ~1 ball)	State 3 (13.4%, ~1 ball)	State 4 (5.3%, ~1 ball)
Dominant clusters	O1 Angle-in (60%), O0 Good swinger (31%), O2 Bouncer (9%)	O0 Good swinger (62%), O2 Bouncer (38%)	O0 Good swinger (70%), O1 Angle-in (24%)	O0 Good swinger (85%), O2 Bouncer (10%)	O1 Angle-in (90%), O0 Good swinger (10%)
Outcome	Dot 78%, Runs 22%, Wicket 0%	Dot 59%, Runs 41%, Wicket 0%	Dot 54%, Runs 44%, Wicket 2%	Dot 51%, Runs 47%, Wicket 2%	Dot 62%, Runs 16%, Wicket 23%
Starts a match in this state	13 / 15	0 / 15	0 / 15	0 / 15	0 / 15

Cummins’ over/round split barely matters here too (round-the-wicket is only 3 balls all dataset, so all 5 states are essentially “over the wicket”). **State 0 is by far his default mode** (58.7%, ~10-ball dwell, angle-in heavy, Dot 78%, Wicket 0%) and is how 13/15 matches begin — this matches the “containment” state from the 2-state model. The new structure is a **four-state escalation cycle** the model decodes once it leaves State 0: State 0 $\square$ State 4 (10.1% of the time) $\square$ State 2 (almost always) $\square$ State 3 (always) $\square$ back to State 0 (27%) or State 1 (73%) $\square$ State 2 (always) $\square$... Through States 1 $\square$ 2 $\square$ 3, the cluster mix shifts steadily from swinger/bouncer towards almost pure good-swinger, and the Dot rate steadily falls (59% $\square$ 54% $\square$ 51%) while Runs rises — a gradual “loosening” similar to the old State 1, but now resolved into discrete steps. **State 4 is the standout: just 5.3% of deliveries (~20 balls), almost entirely the angle-in ball (90%), with a 23% wicket rate** — by far the highest of any state for any bowler in this analysis. It sits right after State 0 in the cycle (State 0 $\square$ State 4 10% of the time, State 4 $\square$

State 2 99.9% of the time), so it reads as a short, sharp “strike” sub-mode breaking out of the containment phase. With ~20 balls behind it, this is the clearest “5-state-only” finding — but also the one most in need of more data before drawing conclusions.

How does the strategy mix change match-by-match?

With each match treated as its own sequence, this plot is now a direct readout of the model’s per-match behaviour rather than an artefact of one long concatenated decode.



Figure 5: Hidden-state mix per match

Each bar is one match (in chronological order, left to right), showing what fraction of that bowler’s deliveries in that match were decoded as State 0 (blue), State 1 (orange), State 2 (red), State 3 (purple) or State 4 (green). The grey number above each bar is how many deliveries that bowler bowled at this batter in that match — many matches are single-figure samples, so individual bars should be read cautiously, but the overall pattern across many matches is informative.

- **M Morkel:** most matches are dominated by a mix of the round-the-wicket states (0/blue, 1/orange, 3/purple), with State 2 (over-the-wicket, red) appearing as a smaller recurring component in many matches rather than a separate match-level mode — consistent with **wicket-position choice being a within-match tactical variable** for Morkel, not just a between-match one. The green State 4 (his 18%-wicket state) appears as thin slivers in roughly a third of matches.
- **SCJ Broad:** most matches show a substantial mix of his round-the-wicket states (0/blue, 1/orange, 3/purple, 4/green), with State 2 (over-the-wicket, red) appearing as a sizeable or even dominant component in about a third of matches. The green State 4 — his highest-wicket state — appears in noticeable amounts across

many matches rather than being concentrated in just one or two, supporting the read that it's a real recurring sub-mode rather than a one-off.

- **MA Starc:** State 1 (orange, the near-absorbing broad mix) and State 0 (blue, tight in-swinger) trade off as the dominant component match-to-match — visually similar to the 2-state picture — but now States 2/3 (red/purple, the alternating finer-grained mixes) also appear as sizeable chunks in many matches. The green State 4 shows up as a small slice in a few matches, and **one whole short match (6 balls) is decoded entirely as State 4** — exactly the kind of match a 2-state model would have folded into “the broad mix” without comment.
- **PJ Cummins:** State 0 (blue, containment) is the largest component in almost every match, consistent with it being the default starting state. States 1–3 (orange/red/purple, the escalation cycle) appear in varying proportions match-to-match. The green State 4 (his 23%-wicket “strike” state) is visible in roughly half of matches, usually as a small slice — consistent with it being a short, occasional sub-mode rather than something Cummins settles into for long.

Home vs away

The dataset records whether the bowling team was playing at home, away, or (rarely) on neutral ground. This field is blank for all the England-vs-Australia (Ashes) matches, so for those we derive it from `Ground Country`: a ground in England or Wales counts as “home” for England and “away” for Australia, and vice versa for grounds in Australia.

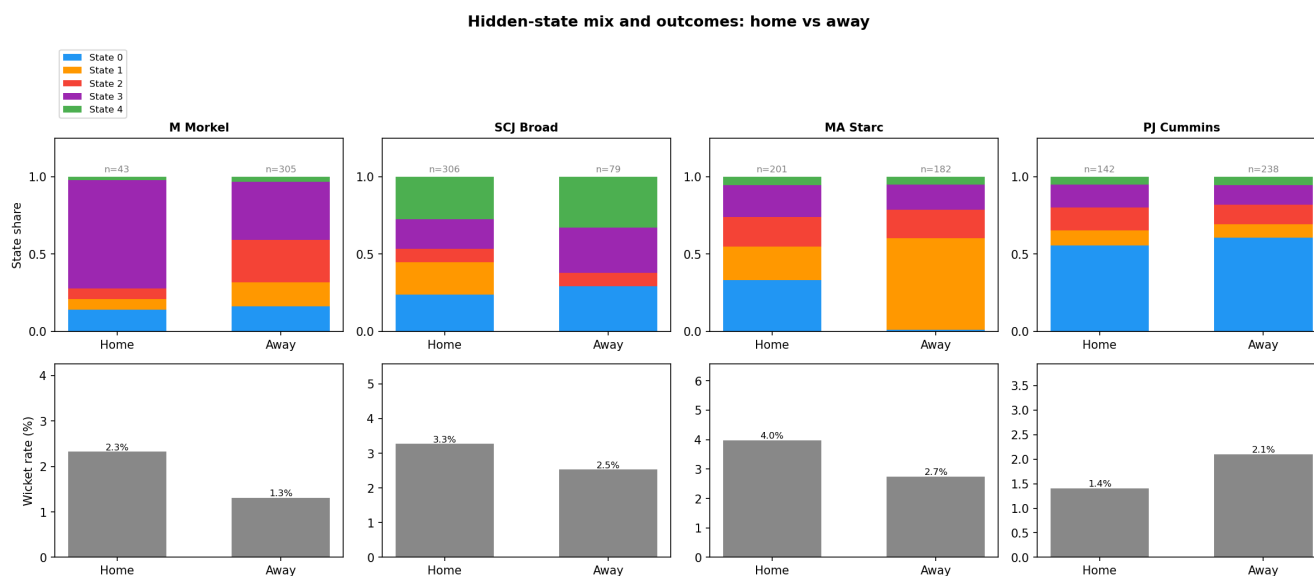


Figure 6: State mix and outcomes: home vs away

The top row shows the State 0–4 split for home vs away deliveries; the bottom row shows the overall wicket rate for home vs away (not split by state, since the per-state-per-venue samples get very small).

- **Wicket rate is higher at home for three of the four bowlers** (Morkel 2.3% vs 1.3%, Broad 3.3% vs 2.5%, Starc 4.0% vs 2.7%; Cummins is the exception at 1.4% home vs 2.1% away). This matches the general expectation that home conditions and crowd support tend to favour the bowling side.
- **Morkel’s home sample (n=43, essentially one match) is dominated by State 3** (70% — his “leakiest” round-the-wicket mode) **vs only 37% away**, where the remaining states (0, 1, 2) are each better represented. The 18%-wicket State 4 is a similarly tiny sliver both home (2%) and away (3%).
- **Broad’s highest-wicket state (State 4) is actually slightly more common away (33%) than at home (27%)**, yet his wicket rate is higher at home (3.3% vs 2.5%) — so, as with the 2-state version, the home/away wicket-rate gap doesn’t come from spending more time in the high-wicket state; something else about home conditions is driving it.
- **Starc’s tight in-swinger opening (State 0) is much more common at home** (33% vs 1% away), while

away matches are dominated by the near-absorbing broad mix (State 1: 59% vs 21% home) — yet the *home* wicket rate is higher (4.0% vs 2.7%). Spending less time in the “broad mix” at home doesn’t reduce his home wicket rate, reinforcing that the home/away gap is more about conditions than which state he’s in.

- **Cummins’ state mix is fairly similar home and away** (State 0 dominant in both: 56% home vs 61% away; State 4, his 23%-wicket state, is 5% in both) — yet his *away* wicket rate is higher (2.1% vs 1.4%). With an almost identical state mix, the away wicket-rate edge must come from within-state differences rather than which “strategy” he’s in.
-

Caveats

- **5 states is a deliberate over-fit, by choice.** BIC prefers 2 states for every bowler, by a wide margin (e.g. Morkel: 1556.8 at 2 states vs 1896.2 at 5). The 2-state model is the statistically “safer” choice, but it mostly just recovers the over/round-the-wicket split. 5 states was chosen here to dig for finer structure underneath that split — some of it (e.g. the small high-wicket states for Morkel, Broad, Starc and Cummins) is genuinely interesting, but it should be treated as **hypothesis-generating, not confirmed** — especially the states occupying under ~10% of deliveries (often only 10–20 balls), which a model with this many parameters can fit to noise.
- **Sample size:** 350–560 balls per bowler, split across 12–21 matches, is small for HMM fitting — many matches contribute only a handful of observations to the per-match decoding, and this gets more acute the more states are added.
- **The “strategies reset each match” assumption is itself a modelling choice**, not a certainty — international bowlers do carry plans and form across a series (e.g. “this batter struggled against X last time, do it again”). Treating matches as fully independent is a reasonable simplification, but match-to-match continuity is not literally zero.
- **The states are descriptive, not causal.** The HMM finds statistical regimes in the sequence; it doesn’t know *why* the regime changed (new ball, declaration tactics, a different batter on strike, fatigue, etc.).
- **Label switching:** hidden states from EM have no inherent ordering. States were sorted by an “aggression score” (wicket rate $\times 10$ + runs rate) purely so State 0...State 4 mean roughly the same thing (low to high aggression) across bowlers, but the absolute values are not directly comparable across bowlers (each HMM was fit independently).
- **Shared clustering with `analysis.md`:** the ball clusters feeding the HMM are identical to the over/round-the-wicket-split clusters described and named in `analysis.md`, so cluster names (O0, R1, ...) mean the same thing in both documents.
- **Starc’s O0 (Length, out-swing)** contains a small number of deliveries with an anomalous (physically implausible) positive Drop Angle — a Hawk-Eye tracking artefact flagged in `analysis.md`, not a genuine delivery type. It sits inside State 4 (5.2% of deliveries) and shouldn’t be over-interpreted.